Reg. No. | | | | | | | | | | | [12AE74]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER – 2015

AEROSPACE DESIGN-I

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	**a.**	Explain the types of aeroplanes depending on their applications.	**04**
	b.	Describe different types of vertical wing positions on the fuselage and what their relative advantages are?	**05**
	c.	Explain the motor systems of the conventional helicopter.	**05**
2.	**a.**	Explain monocoque and semi monocoque structure and their applications.	**06**
	b.	Briefly explain structural sub assembly of a transport aeroplane.	**04**
	c.	Explain the structural materials used for the construction of wing, fuselage and tail planes.	**04**
3.	**a.**	Derive the expression for the banking angle of the aircraft wing during a turn in the horizontal plane. Velocity of the aircraft=V_m/sec and radius of turn=R.	**04**
	b.	A trainer aircraft is maneuvering in the vertical plane in a circular path at velocity=480Kmph. What is the permissible minimum radius of the path if the load factor should not exceed 5. What is the minimum load factor experienced by the aircraft in its circular path?	**06**
	c.	Derive the expression for the load factor experienced by an aircraft in level flight at velocity V_m/sec when it encounters a sudden vertical gust of velocity U_m/sec.	**04**
4.	**a.**	Explain distortion energy theory and maximum stress theory.	**06**
	b.	A thin walled circular cylinder has a wall thickness=2mm and inner diameter=250mm. its ends are closed and it contains fluid under pressure=PN/sqmm. The cylinder is subjected to an axial compressive load=50,000N. Yield strength of the Material=400N/sqmm. Compute the maximum internal pressure permissible based on distortion energy theory of yielding.	**08**
5.	**a.**	Explain buckling of the flat panel supported on all four sides and buckling coefficient in compression stress.	**04**
	b.	Explain effective width under compressive stress.	**04**

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c. Compute the initial buckling stress for an Al-alloy rectangular section shown in **Fig.1** Young's modus for al-alloy=72000N/sqmm. **06**

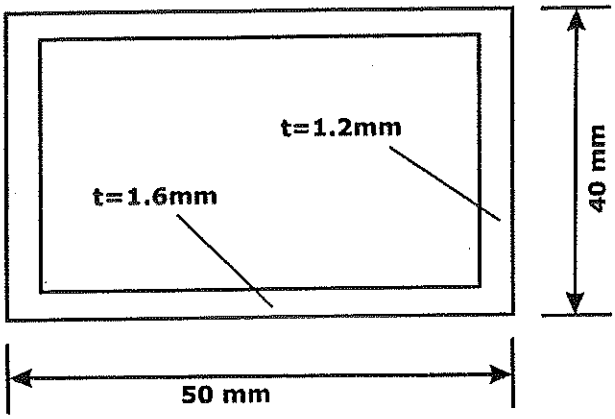

Figure.1

6. **a.** Sketch the planform and cross section of a two spar using and explain its components. **06**

b. Compute the shear flow in the box beam shown in Fig-2. Assume that the web is not effective in bending. Areas of flange members are a_1=80mm^2, a_2=60mm^2, a_3=100mm^2 and V=1000N. **08**

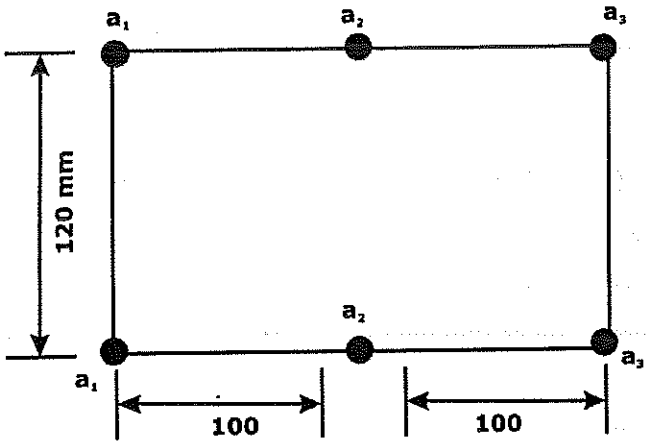

Figure 2

7. **a.** Explain the functions and forces on landing gear. **05**

b. Explain limit load, margin of safety and design load on a component. **03**

c. Compute the margin of safety in a stepped bar member shown in Fig-3. Ultimate strength of the material=800N/sqmm static stress concentration factor=1.6, Force along the axis=40000N (limit), Factor of safety to be used=1.5. **06**

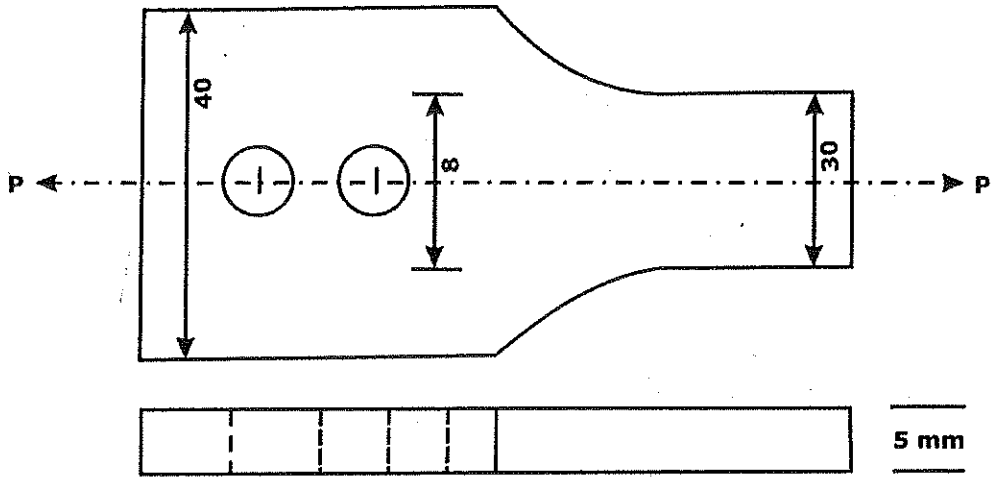

Figure 3

8. Write short note on any four: **14**

i. Design requirements of structure.

ii. Bolted joint.

iii. Forces on tail plane.

iv. Solid and blind rivets.

v. Principal axis of inertia of the cross section.

vi. Flying characteristics of a helicopter.